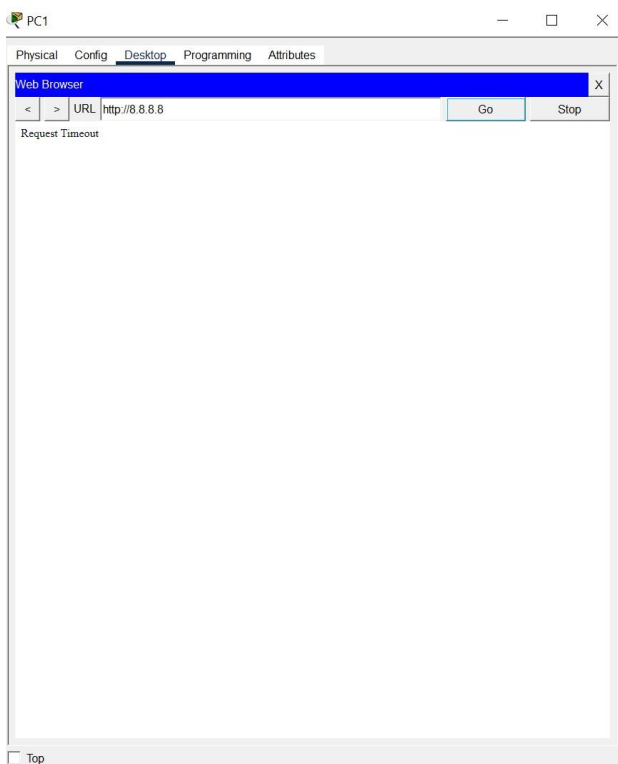


Section 1: Troubleshooting Stage

Screenshot 1 Location:

- **Section Title:** 1. Documenting the Initial Web Browser Connection Failure
- **Details:** This screenshot documents the initial failure when attempting to access the HTTP web server (8.8.8.8) from PC1, resulting in a 'Request Timeout' error. This issue was caused by an incorrectly structured Extended Access List (ACL 100) on Router R1 that implicitly dropped return traffic, combined with missing default gateway parameters and reverse path routing configurations across the topology.



Screenshot 2 Location:

- **Section Title:** 2. Verifying NAT Translations on Router R1 Before Remediation
- **Details:**

This capture shows the verification attempt of the NAT translation table using the **show ip nat translations** command on Router R1 prior to applying fixes. No dynamic translations were recorded because Access List 1 was incorrectly configured to permit a non-existent network (192.168.50.0), and the PAT overload process was erroneously mapped to an inside interface instead of the outside exit interface.

The screenshot shows a Cisco Router CLI window titled 'Router1'. The window has tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The user has entered several commands to configure NAT and access lists. The output shows that NAT is configured with a static translation for 10.0.0.3 to 192.168.10.10. The access list 100 is configured to permit TCP traffic from 192.168.10.0/24 to 8.8.8.8. The access list 10 is configured to deny host 192.168.20.10 and permit any other traffic. The NAT statistics show 1 total translation, 1 static, 0 dynamic, and 0 extended. The configuration is saved to the startup configuration.

```
Router(config)#do show ip nat statistics
show ip nt statistics
^
% Invalid input detected at '^' marker.

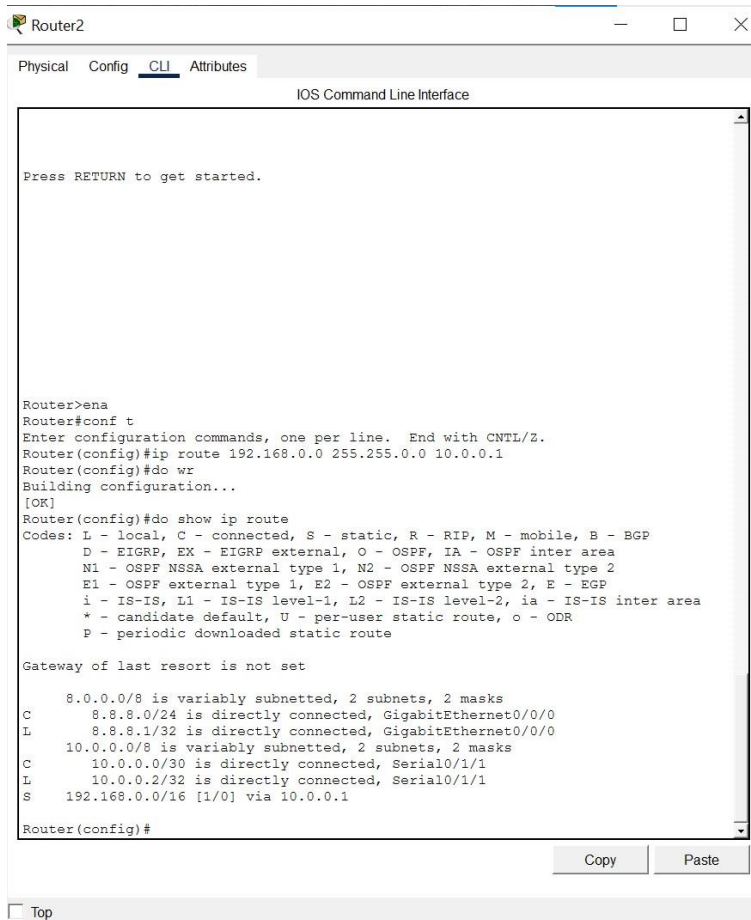
Router(config)#do show ip nat statistics
Total translations: 1 (1 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial0/1/1
Inside Interfaces: GigabitEthernet0/0/0 , GigabitEthernet0/0/1
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router(config)#do wr
Building configuration...
[OK]
Router(config)#interface Serial0/1/1
Router(config-if)# no ip access-group 100 out
Router(config-if)#exit
Router(config)#no access-list 100
Router(config)#access-list 100 permit tcp 192.168.10.0 0.0.0.255 host 8.8.8.8 eq
80
Router(config)#interface Gig0/0/0
Router(config-if)# ip access-group 100 in
Router(config-if)#exit
Router(config)#access-list 10 deny host 192.168.20.10
Router(config)#access-list 10 permit any
Router(config)#interface Gig0/0/1
Router(config-if)# ip access-group 10 in
Router(config-if)#end
Router#
%SYS-S-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#do show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
---  ---
tcp  10.0.0.3:1027         192.168.10.10:1027 8.8.8.8:80         8.8.8.8:80
tcp  10.0.0.3:1028         192.168.10.10:1028 8.8.8.8:80         8.8.8.8:80
Router(config)#
```

Screenshot 4 Location:

- **Section Title:** 4. Verifying the Routing Table and Return Path on Router R2
- **Details:**

This output confirms the correction of the routing parameters on Router R2. After resolving a syntax placement error where the command was previously trapped inside interface configuration mode, the static route was successfully written globally. The routing table now actively lists the network under code S as **192.168.0.0/16 via 10.0.0.1**, guaranteeing a proper return path for packets.



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.0.0 255.255.0.0 10.0.0.1
Router(config)#do wr
Building configuration...
[OK]
Router(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

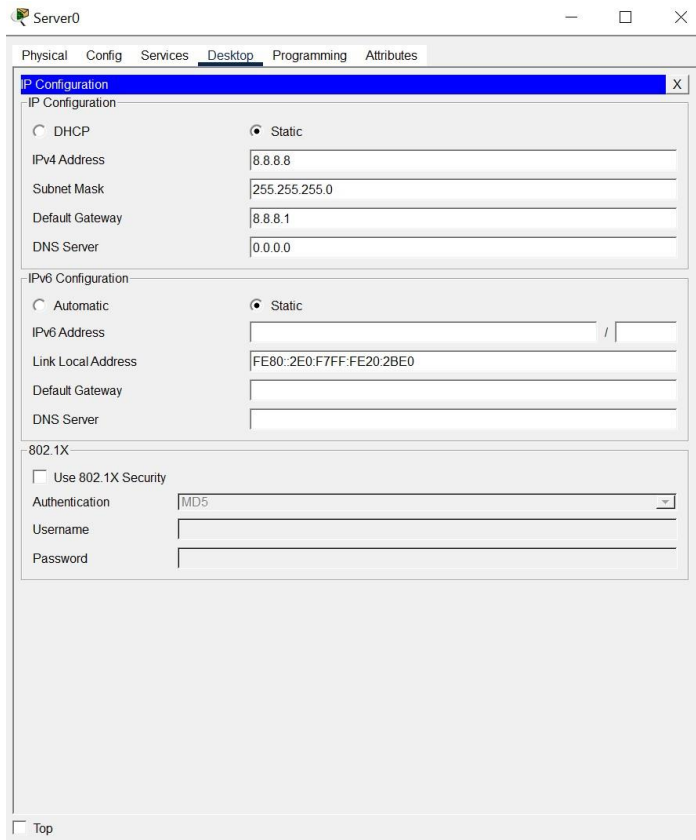
      0.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       8.8.8.0/24 is directly connected, GigabitEthernet0/0/0
L       8.8.8.1/32 is directly connected, GigabitEthernet0/0/0
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/1
L       10.0.0.2/32 is directly connected, Serial0/1/1
S       192.168.0.0/16 [1/0] via 10.0.0.1

Router(config)#
```

Screenshot 5 Location:

- **Section Title:** 5. Correcting Server IP Configuration and Default Gateway Parameters
- **Details:**

This image represents the final resolution of the endpoint infrastructure issue on Server0. During advanced troubleshooting, it was discovered that the server's default gateway was missing (0.0.0.0) and the subnet mask was mismatched. The parameters were updated to Class C (255.255.255.0) and the gateway was pointed to **8.8.8.1**, enabling the server to respond to external requests.



Section 3: Verification Stage

Screenshot 6 Location:

- **Section Title:** 6. Successful End-to-End HTTP Web Service Verification from PC1
- **Details:**

The final successful outcome of the lab layout; PC1 successfully establishes an end-to-end HTTP session with the web server, loading the default Cisco Packet Tracer greeting page instantly. This confirms the flawless operation of PAT overload, stable asymmetric static routing on R2, and the modified ACL rules allowing established TCP return traffic.



Screenshot 7 Location:

- **Section Title:** 7. Validating Unauthorized Host Security Blocks via Standard ACL
- **Details:**

Verification of the second security requirement within the lab criteria. When executing a ICMP ping request from the unauthorized host PC2 to the server address **8.8.8.8**, the packet is immediately dropped at the gateway level, returning a **Destination host unreachable** packet from **192.168.20.1**. This validates that Standard ACL 10 applied inbound on interface **Gig0/0/1** is operational and functioning as intended

